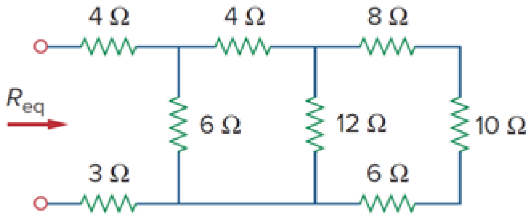
Chapter 2, Problem 9PP

Problem

By combining the resistors in Fig. 2.36, find *R*eq.

Figure 2.36:



Step-by-step solution

Step 1 of 4

Consider the circuit of textbook Figure 2.36. The resistors $8\ \Omega$, $10\ \Omega$ and $6\ \Omega$ are in series. Therefore, their equivalent resistance is,

$$8\ \Omega + 10\ \Omega + 6\ \Omega = 24\ \Omega$$

The resultant diagram is as shown in Figure 1.

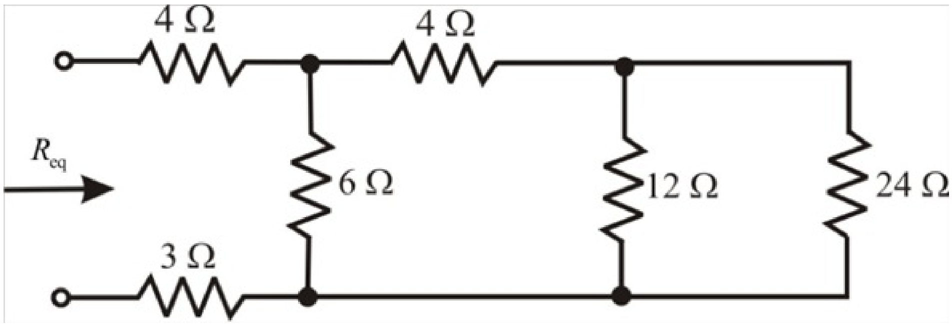


Figure 1

Step 2 of 4

The resistors 24Ω and 12Ω in Figure 1 are in parallel. Therefore their equivalent resistance is,

$$\begin{aligned}(12 \parallel 24) &= \frac{(12)(24)}{36} \\ &= 8\ \Omega\end{aligned}$$

The resultant diagram is as shown in Figure 2.

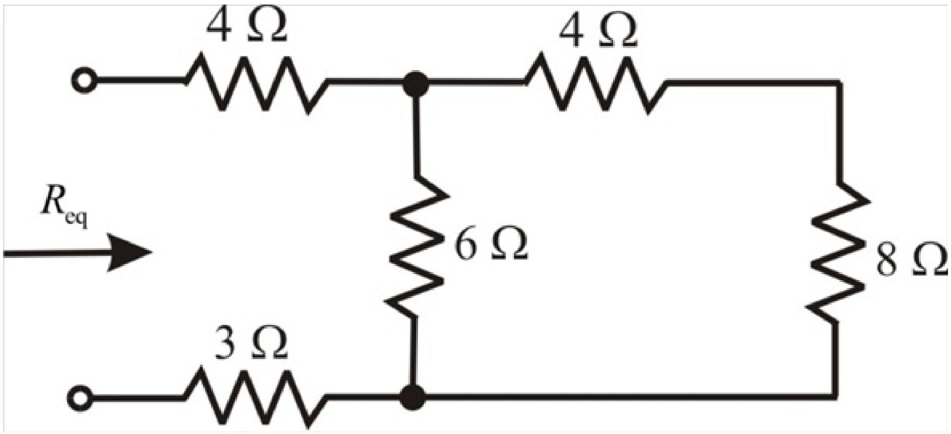


Figure 2

Step 3 of 4

The resistors 4Ω and 8Ω are in series and the resultant is in parallel to 6Ω. Therefore their equivalent resistance is,

$$\begin{aligned}\left[(8+6) \parallel 6\right] &= 12 \parallel 6 \\ &= \frac{(12)(6)}{18} \\ &= 4\ \Omega\end{aligned}$$

The resultant diagram is as shown in Figure 3.

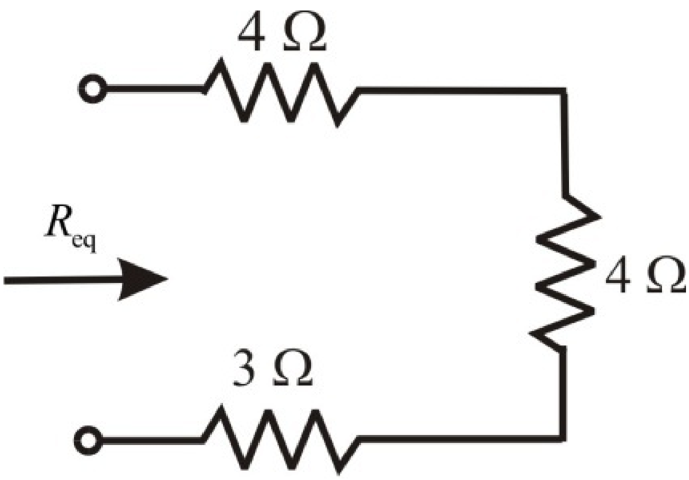


Figure 3

Step 4 of 4

The equivalent resistance of Figure 3 is,

$$\begin{aligned}R_{eq} &= 4+4+3 \\ &= 11\ \Omega\end{aligned}$$

Therefore, the equivalent resistance of the textbook Figure 2.36 is $\boxed{R_{eq} = 11\ \Omega}$.